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ORIGINAL SCIENTIFIC PAPER
Arrhythmias

A practical 5-stage clinical scale for electrical storm: the STORM classification

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Aims Electrical storm (ES) is a highly heterogeneous condition with wide-ranging clinical presentations. The absence of standardized classification hampers risk stratification and limits effective multidisciplinary coordination.

Objective The aim of this study was to develop a classification system based on simple clinical characteristics and stratify 30-day mortality in patients with ES.

Methods and results Patients admitted to intensive care units for ES between 2010 and 2023 across four tertiary centres were retrospectively included. The five-stage STORM severity-response classification, based on treatment intensity during hospitalization, incorporated four clinically relevant parameters: signs of acute heart failure or haemodynamic instability, need for inotropes or vasopressors, use of advanced supportive therapies (including deep sedation) and renal replacement therapy, and implementation of mechanical circulatory support. The primary outcome was all-cause mortality at 30 days. A total of 788 patients were included. The cohort was predominantly male (84.3%), with a median age of 66.0 years (57.0–74.0). The majority had ischaemic cardiomyopathy (65.6%), with a median LVEF of 30.0% (20.0–45.0). According to the STORM classification, 421 patients (53.4%) were categorized as STORM-1, 45 (5.7%) as STORM-2, 86 (10.9%) as STORM-3, 220 (27.9%) as STORM-4 and 16 (2.0%) as STORM-5. Overall, 117 patients (14.8%) died within 30 days. A stepwise increase in 30-day mortality was observed across STORM stages – 5.0%, 6.7%, 20.9%, 30.5% and 50.0% for stages 1 through 5, respectively ($P < 0.01$).

Conclusion The STORM classification may facilitate standardized multidisciplinary management strategies and effectively stratifies 30-day mortality risk in patients with ES, ranging from 5% in stage 1% to 50% in stage 5. Further prospective studies are warranted to validate our findings.

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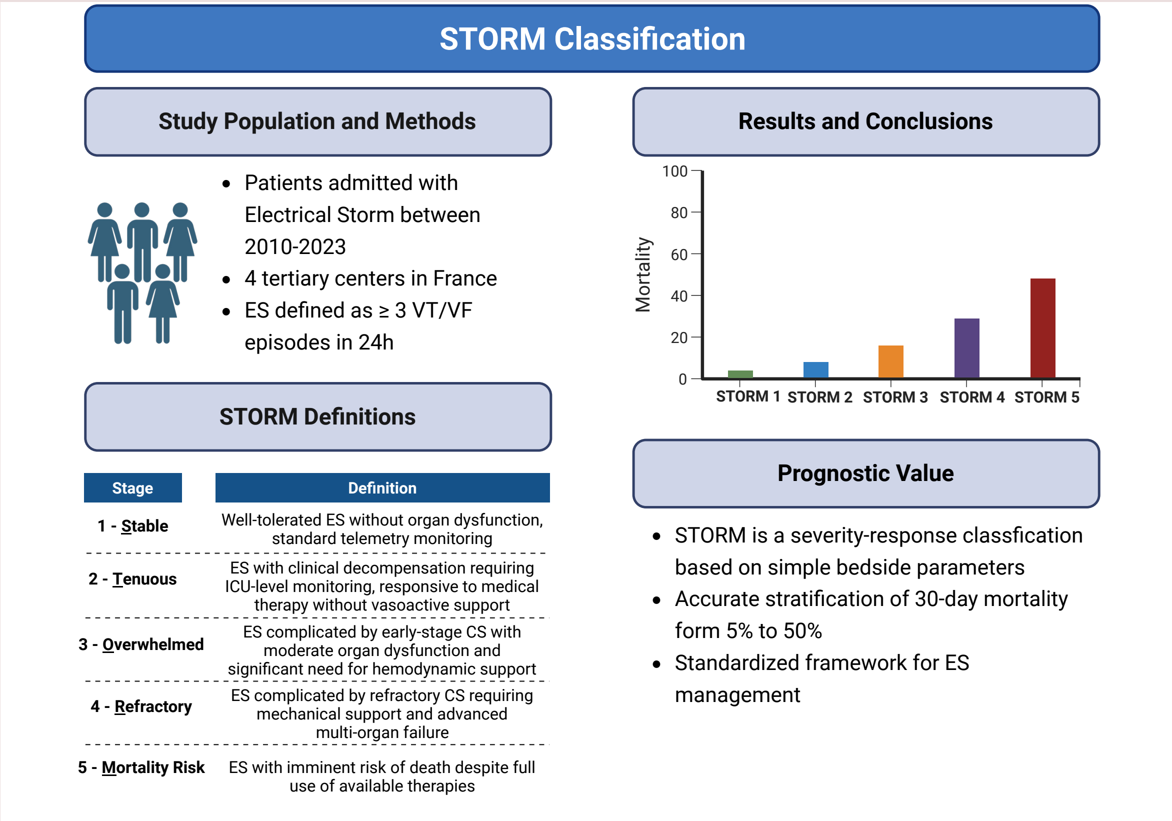
Graphical abstract



Keywords Electrical storm • Mortality • Risk stratification • Classification • Ventricular arrhythmia • Ventricular tachycardia • Ventricular fibrillation

Introduction

Electrical Storm (ES) is characterized by the occurrence of three or more episodes of sustained ventricular arrhythmias (VAs) within 24 h requiring termination by antitachycardia pacing (ATP) or electrical shock, each event separated by at least 5 min.[1] In recent years, therapeutic options for ES have significantly improved and diversified, both from an electrophysiological standpoint, including antiarrhythmic drugs, catheter ablation and neuraxial modulation (such as cardiac sympathetic denervation and stellate ganglion blockade), and from a haemodynamic perspective, with the advent of advanced pharmacological and device-based therapies.[2] Yet, the mortality associated with ES remains unacceptably high, reaching up to 34% at one year,[3] and continues to represent a major challenge for the medical community. One possible explanation lies in the marked heterogeneity of the phenotypes and underlying aetiologies observed in ES, each requiring a tailored management approach.[4]

The clinical expression of ES is markedly variable, extending from asymptomatic episodes controlled by ATP to severe, drug-refractory arrhythmias associated with acute haemodynamic deterioration or cardiac arrest.[5] Although these diverse scenarios all meet the current definition of ES, their implications in terms of morbidity and mortality differ markedly. Jentzer et al.[4] emphasized the need to stratify ES into different severity stages to support a more rational and targeted escalation of therapy. More recently, a clinical consensus statement from the European Heart Rhythm Association – endorsed by the Asia-Pacific Heart Rhythm Society, Heart Rhythm Society and Latin American Heart Rhythm Society[5] – proposed a stepwise management algorithm grounded in simple clinical parameters to guide treatment decisions based on ES severity.

However, despite guideline recommendations emphasizing the importance of haemodynamic assessment and risk stratification in ES management,[4,5] a standardized and universally accepted classification system for stratifying the severity of ES remains lacking, and studies conducted in this field continue to be hampered by substantial heterogeneity. In this context, developing a robust and clinically meaningful classification framework is essential – not only to improve decision-making and risk stratification, but also to facilitate communication among multidisciplinary teams, including cardiologists, electrophysiologists, anaesthesiologists and cardiac surgeons.[6–8] Therefore, the present study aimed to propose a simple classification system based on clinically relevant and readily accessible bedside data to stratify 30-day mortality risk in patients presenting with ES.

Methods

Study patients

We conducted a retrospective multicentre study involving patients admitted to intensive care units in four tertiary centres in France (Dijon, Lille, Rennes and St-Etienne) who experienced an ES between January 2010 and March 2023.[9,10] ES was defined as the occurrence of at least three sustained episodes of ventricular tachycardia (VT) or ventricular fibrillation (VF) within 24 h, each separated by at least 5 min. Patients with structural heart disease were included, as were those without structural heart disease such as channelopathy, drug-induced torsades de pointes, or idiopathic VA. Patients under 18 years of age or under legal protection were excluded.

Data collection

Various data were collected, including demographic characteristics, underlying cardiac disease, left ventricular ejection fraction (LVEF), presence of an implantable cardioverter-defibrillator (ICD), medical history, chronic medications, potential triggers of ES (notably ST-elevation myocardial infarction [STEMI]), ES management strategies and in-hospital complications. All data were retrospectively collected using electronic data files and laboratory data. Regarding the type of cardiomyopathy, patients were defined as having ischaemic, non-ischaemic or 'other' cardiomyopathies (including hypertrophic, valvular, congenital, peripartum cardiomyopathies, myocarditis as well as non-structural heart diseases such as long QT syndrome or Brugada syndrome). Patients were considered to have had a history of VA ablation if they underwent catheter ablation or stereotactic ablation. Acute heart failure was defined as a clinical syndrome with rapidly developing or worsening typical heart failure symptoms and signs requiring emergency treatment, characterized by left-sided signs (dyspnoea, rales and crepitations) and/or right-sided signs (ascites, hepatojugular reflux, hepatomegaly, lower limb oedema, pleural effusion).[11] Severe haemodynamic instability was defined as systolic blood pressure (SBP) < 90 mmHg despite adequate fluid resuscitation with clinical and laboratory evidence of end-organ dysfunction, or the need for vasopressors to maintain SBP ≥90 mmHg.

Ethics

Ethical approval was not sought for the present study according to institutional policies regarding retrospective studies. This study adhered to the Declaration of Helsinki guidelines for ethical research.

STORM classification

The STORM classification is designed as a severity-response framework that reflects both disease severity and treatment intensity during hospitalization. Based on prior studies, guidelines and expert consensus,[1,3,5,12] four clinically relevant characteristics were analysed for each patient (*Figure 1*): (1) signs suggestive of acute heart failure or haemodynamic instability; (2) the need for inotropes or vasopressors; (3) the requirement for advanced supportive therapies (including deep sedation) or renal replacement therapy; and (4) the use of mechanical circulatory support (MCS) including extracorporeal membrane oxygenation (ECMO), Impella and intra-aortic balloon pump (IABP). Treatment escalation during hospitalization for ES was used as a surrogate marker of persistent hypotension and hypoperfusion, allowing retrospective assessment of maximum clinical deterioration, since haemodynamic and metabolic parameters were only recorded at admission. Based on these four characteristics, a 5-stages classification was developed (*Figure 2*) and each patient included in the analysis was staged as follows:

- Stage 1 (Stable) refers to well-tolerated episodes of ES without acute heart failure or haemodynamic instability, managed with standard telemetry monitoring in a cardiac care unit
- Stage 2 (Tenuous) corresponds to ES with acute heart failure or haemodynamic instability – defined by systolic blood pressure ≤90 mmHg, lactate >2 mmol/L, or oliguria – that responds to first-line medical therapy (diuretics for fluid overload, fluid resuscitation for hypovolemia) without requiring vasopressor or inotropic support. These patients require ICU-level monitoring including continuous arterial line, frequent vital signs assessment and repeated urine output measurement to closely track clinical response and anticipate potential therapeutic escalation.
- Stage 3 (Overwhelmed) indicates ES with haemodynamic instability persisting despite first-line medical management, requiring initiation of inotropes or vasopressors, or necessitating advanced supportive therapies (deep sedation for refractory arrhythmias) or renal replacement therapy, but without need for mechanical circulatory support.
- Stage 4 (Refractory) denotes ES with severe cardiogenic shock requiring mechanical circulatory support (ECMO, Impella, or IABP) in conjunction with inotropic/vasopressor therapy and associated with multi-organ failure.
- Stage 5 (Mortality Risk) designates ES with refractory cardiogenic shock requiring escalating mechanical and pharmacological support with imminent risk of death despite maximal therapeutic interventions.

Outcomes

The primary outcome was 30-day all-cause mortality across STORM stages. Secondary outcomes included in-hospital complications and VA recurrence, cardiovascular and non-cardiovascular mortality and the number of left ventricular assist device (LVAD) implantations or heart transplantations performed.

Statistical analysis

Continuous variables were reported as medians and interquartile ranges (IQR). Categorical variables were described as frequencies and percentages. Quantitative variables were compared using the Kruskal-Wallis test for multiple groups. Categorical variables were compared using the Pearson chi-square test or the Fisher exact test when appropriate. To analyse trends across quartiles, we employed the Cochran-Armitage test for qualitative variables and the Jonckheere-Terpstra test for quantitative variables, both described in the text and tables as '*P*-trend'. The primary outcome of 30-day all-cause mortality was visualized using Kaplan–Meier cumulative event curves and differences between STORM stages were assessed using the log-rank test. To address potential immortal time bias, a landmark analysis was conducted on day 7, restricted to patients who were alive at that time. To ensure the robustness of the findings and account for potential confounding, a multivariable logistic regression analysis was also performed as a sensitivity analysis to assess the association between STORM stages and 30-day all-cause mortality. Both Cox regression and logistic regression models were adjusted for the following variables: age, sex, obesity, type of cardiomyopathy (ischaemic, dilated, other), history of cardiac surgery or atrial fibrillation, cardiovascular risk factors, comorbidities (peripheral artery disease and chronic kidney disease), baseline antiarrhythmic and heart failure therapies prior to ES onset, history of VA and ICD implantation before ES onset, inaugural presentation of ES, STEMI as an acute trigger, acute treatment with amiodarone or lidocaine and catheter ablation. Results are presented as adjusted hazard ratios (aHRs) and adjusted odds ratios (aORs) with 95% confidence intervals (CIs). Eventually, building on gaps identified in literature and relevant pathophysiological considerations,[1,2,4] we conducted prespecified subgroup analyses to assess whether the impact of STORM stages differed according to ischaemic cardiomyopathy, LVEF ≤ 35%, an AMI-triggered ES or the absence of an identifiable trigger, and whether presentation occurred as monomorphic VT or polymorphic VT/VF. All tests were two-tailed. The value of $P \leq 0.05$ was accepted as statistically significant. Analyses were performed using R software [version 4.3.2 (2023-10-31)].

Results

Baseline characteristics

As shown in *Figure 3*, the analysis included 788 patients with ES, after exclusion of 19 patients due to missing data. Based on the STORM classification, 421 patients (53.4%) were classified as STORM-1, 45 (5.7%) as STORM-2, 86 (10.9%) as STORM-3, 220 (27.9%) as STORM-4 and 16 (2.0%) as STORM-5. *Table 1* summarizes baseline characteristics. The median age was 66.0 years (57.0–74.0) and 84.3% of patients

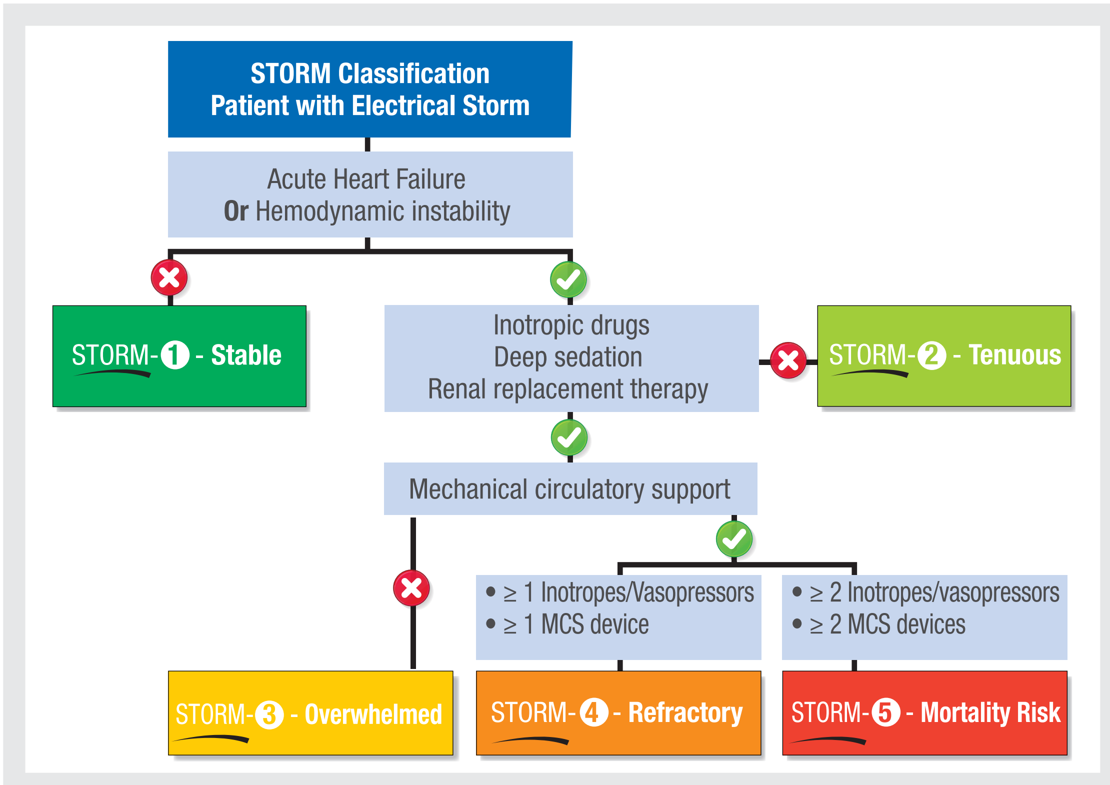


Figure 1 STORM classification algorithm for risk stratification in electrical storm patients. MCS, mechanical circulatory support.

were male. Overall, 517 patients (65.6%) had a history of ischaemic cardiomyopathy, while 143 (18.1%) presented with dilated cardiomyopathy. Approximately half of the cohort ($n = 388$, 49.2%) had previously experienced VA and 453 patients (57.5%) were equipped with an ICD.

STORM-5 patients were younger (54.0 vs. 67.0 years for STORM-1, $P < 0.01$; P-trend < 0.01) and demonstrated a higher prevalence of history of ischaemic cardiomyopathy (87.5% vs. 61.5%, $P = 0.04$), along with significantly lower baseline LVEF (25.0% vs. 35.0%, $P < 0.01$), with consistent trends across risk categories (P-trend < 0.01). They also exhibited fewer comorbidities, including atrial fibrillation and peripheral artery disease (P-trend < 0.01). In addition, they showed a markedly lower prevalence of prior VA (6.2% vs. 64.1%, $P < 0.01$), previous VA ablation (0.0% vs. 20.2%, $P < 0.01$) and ICD implantation (6.2% vs. 72.0%, $P < 0.01$). Conversely, they more frequently presented with ES as the initial manifestation of their underlying cardiomyopathy (31.2% vs. 10.0%, $P < 0.01$).

ES features and in-hospital management

The rate of VF demonstrated an increasing trend across STORM stages (P-trend < 0.01), rising from 27.3% in STORM-1 to 57.1% in STORM-5 ($P < 0.01$). Similarly, the proportion of ESs triggered by STEMI increased significantly with STORM severity, reaching 81.2% in STORM-5 compared to 5.9% in STORM-1 ($P < 0.01$, P-trend < 0.01) (*Table 2*). The use of amiodarone rose from 78.4% in STORM-1 to 93.8% in STORM-5 ($P < 0.01$, P-trend < 0.01), while lidocaine increased from 29.0% to 87.5% ($P < 0.01$, P-trend < 0.01) (*Table 3*). Catheter ablation was less frequently used in higher-risk categories (43.2% in STORM-1 vs. 6.2% in STORM-5, P-trend < 0.01). Conversely, neuraxial modulation was more frequently implemented in advanced stages (25.0% in STORM-5 vs. 1.0% in STORM-1, P-trend < 0.01). From STORM-3 to STORM-5, the use of individual inotropic and vasopressor agents increased significantly, with dobutamine rising from 37.2% to 93.8%, norepinephrine from 52.3% to 87.5% and epinephrine from 10.5% to 50.0% (all $P < 0.01$, P-trend < 0.01). ECMO was the most frequently utilized modality for temporary MCS, administered in 80 patients (10.2%), including 65 in STORM-4 (29.5%) and 15 in STORM-5 (93.8%). IABP and Impella were used in 21 (2.7%) and 12 (1.5%) patients, respectively. Deep sedation was implemented in 233 patients (29.6%), predominantly in STORM-4 (175, 79.5%) and STORM-5 (14, 87.5%). Renal replacement therapy was required in 36 patients (4.6%), with 28 in STORM-4 (12.7%) and 5 in STORM-5 (31.2%).

30-day all-cause mortality and secondary outcomes

Overall, 117 patients (14.8%) had died at 30 days, with mortality rates increasing significantly across STORM stages: 5.0% in STORM-1 (21 patients), 6.7% in STORM-2 (3 patients), 20.9% in STORM-3 (18 patients), 30.5% in STORM-4 (67 patients) and 50.0% in STORM-5 (8 patients) ($P < 0.01$) (*Figure 4A*). Using STORM-1 as the reference, the adjusted HRs for 30-day all-cause mortality showed a significant stepwise increase across STORM stages: STORM-2: aHR 1.11 (0.33–3.79, $P = 0.87$); STORM-3: aHR 3.96 (2.02–7.71, $P < 0.01$); STORM-4: aHR 6.62 (3.79–11.57, $P < 0.01$); and STORM-5: aHR 14.36 (5.26–19.51, $P < 0.01$). In the landmark analysis, the progressive increase in mortality risk across STORM stages persisted, with aHRs ranging from 3.10 (1.29–7.47) for STORM-3 to 8.35 (6.01–10.74) for STORM-5

STORM-1	Stable	• **Clinical signs:** No acute heart failure, no hemodynamic instability (SBP ≥90 mmHg) • **Clinical tolerance:** No organ failure, no need for vasopressor • **Summary:** Well-tolerated ES without organ dysfunction, standard telemetry monitoring
STORM-2	Tenuous	• **Clinical signs:** Acute heart failure or hemodynamic instability (SBP ≤90 mmHg, lactate >2 mmol/L, or oliguria) responsive to first-line therapy • **Clinical tolerance:** Reversible organ dysfunction without need for vasopressors or inotropes • **Summary:** ES with clinical decompensation requiring ICU-level monitoring (arterial line, frequent vitals), responsive to medical therapy without vasoactive support
STORM-3	Overwhelmed	• **Clinical signs:** Significant hypotension (SBP ≤90 mmHg) requiring moderate doses of inotropes/vasopressors • **Clinical tolerance:** Moderate organ dysfunction (elevated lactate levels, moderate renal impairment), without critical failure • **Summary:** ES complicated by early-stage CS with moderate organ dysfunction and significant need for hemodynamic support
STORM-4	Refractory	• **Clinical signs:** Severe CS requiring inotropes/vasopressors and mechanical circulatory support (ECMO, Impella, IABP) • **Clinical tolerance:** Severe multi-organ failure (renal, hepatic, respiratory) • **Summary:** ES complicated by refractory CS requiring mechanical support and advanced multi-organ failure
STORM-5	Mortality Risk	• **Clinical signs:** End-stage CS with persistent hemodynamic instability despite maximal mechanical and pharmacological support • **Clinical tolerance:** Multi-organ failure with a severely limited prognosis • **Summary:** ES with imminent risk of death despite full use of available therapies

Figure 2 STORM classification: clinical characteristics and severity stages for electrical storm patients. ECMO, extracorporeal membrane oxygenation; ES, electrical storm; IABP, intra-aortic balloon pump; SBP, systolic blood pressure.

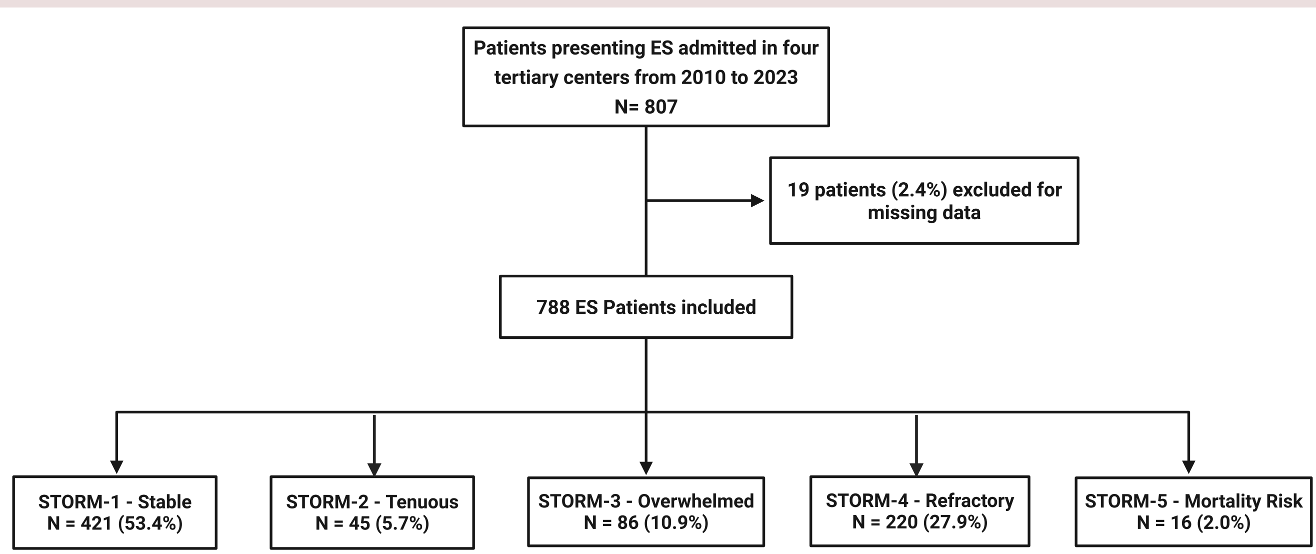


Figure 3 Flow chart of the study. ES, electrical storm.

Table 1 Baseline characteristics according to the STORM classification[a]

	Overall population (*n* = 788)	STORM 1 (*n* = 421)	STORM 2 (*n* = 45)	STORM 3 (*n* = 86)	STORM 4 (*n* = 220)	STORM 5 (*n* = 16)	*P* value	*P*-trend
Age, years, median (IQR)	66.0 (57.0–74.0)	67.0 (58.0–76.0)	71.0 (61.0–75.0)	68.0 (56.0–73.8)	64.0 (56.0–71.0)	54.0 (50.8–62.8)	<0.01	<0.01
Male sex, *n* (%)	664 (84.3)	357 (84.8)	31 (68.9)	74 (86.0)	189 (85.9)	13 (81.2)	0.09	0.72
BMI, kg/m^2, median (IQR)	26.8 (24.0–30.1) (*n* = 750)	27.4 (24.5–30.7) (*n* = 409)	26.1 (23.4–29.4) (*n* = 42)	25.6 (23.7–28.7) (*n* = 85)	26.0 (23.4–29.6) (*n* = 199)	27.4 (22.9–33.0) (*n* = 15)	<0.01	<0.01
Cardiomyopathy, *n* (%)							0.04	
Ischaemic	517 (65.6)	259 (61.5)	27 (60.0)	58 (67.4)	159 (72.3)	14 (87.5)		<0.01
Dilated	143 (18.1)	86 (20.4)	12 (26.7)	11 (12.8)	34 (15.5)	0 (0.0)		0.02
Other	128 (16.2)	76 (18.1)	6 (13.3)	17 (19.8)	27 (12.3)	2 (12.5)		0.10
LVEF, %, median (IQR)	30.0 (20.0–45.0)	35.0 (25.0–45.0)	30.0 (22.0–40.0)	30.0 (20.0–45.0)	25.0 (19.8–39.0)	25.0 (15.0–36.0)	<0.01	<0.01
Medical history, *n* (%)								
Cardiac surgery	68 (8.6)	38 (9.0)	6 (13.3)	8 (9.3)	15 (6.8)	1 (6.2)	0.62	0.35
Atrial fibrillation	296 (37.6)	174 (41.3)	23 (51.1)	35 (40.7)	62 (28.2)	2 (12.5)	<0.01	<0.01
Hypertension	417 (52.9)	219 (52.0)	27 (60.0)	48 (55.8)	114 (51.8)	9 (56.2)	0.83	0.88
Diabetes mellitus	207 (26.3)	105 (24.9)	13 (28.9)	22 (25.6)	58 (26.4)	9 (56.2)	0.12	0.25
Smoking	332 (42.1)	168 (39.9)	35 (40.7)	35 (40.7)	104 (47.3)	9 (56.2)	0.25	0.05
>2 cardiovascular risk factors	568 (72.1)	312 (74.1)	31 (68.9)	61 (70.9)	150 (68.2)	14 (87.5)	0.33	0.27
PAD	100 (12.7)	59 (14.0)	8 (17.8)	15 (17.4)	17 (7.7)	1 (6.2)	0.047	0.04
COPD	49 (11.4) (*n* = 428)	29 (12.8) (*n* = 227)	1 (3.3) (*n* = 30)	8 (15.1) (*n* = 53)	11 (10.6) (*n* = 104)	0 (0.0) (*n* = 14)	0.35	0.40
CKD	147 (18.7)	87 (20.7)	9 (20.0)	15 (17.4)	35 (15.9)	1 (6.2)	0.45	0.07
Treatment before ES, *n* (%)								
None	42 (5.3)	11 (2.6)	0 (0.0)	5 (5.8)	22 (10.0)	4 (25.0)	<0.01	<0.01
Amiodarone	209 (26.5)	139 (30.9)	16 (35.6)	25 (29.1)	38 (17.3)	0 (0.0)	<0.01	<0.01
Betablocker	561 (71.2)	342 (81.2)	35 (77.8)	52 (60.5)	123 (55.9)	9 (56.2)	<0.01	<0.01
Flecainide	14 (1.8)	10 (2.4)	0 (0.0)	3 (3.5)	1 (0.5)	0 (0.0)	0.41	0.13
Other antiarrhythmic drug	16 (2.0)	13 (3.1)	0 (0.0)	1 (1.2)	2 (0.9)	0 (0.0)	0.59	0.04
Oral anticoagulation	540 (68.5)	313 (74.3)	30 (66.7)	54 (62.8)	139 (63.2)	4 (25.0)	<0.01	<0.01
Antiplatelet	174 (22.1)	97 (23.0)	16 (35.6)	19 (22.1)	39 (17.7)	3 (18.8)	0.24	0.12
ACE-I of ARB	403 (51.1)	231 (54.9)	25 (55.6)	40 (46.5)	97 (44.1)	10 (62.5)	0.07	0.02
MRA	216 (27.4)	127 (30.2)	11 (24.4)	25 (29.1)	52 (23.6)	1 (6.2)	0.24	0.03
Valsartan-Sacubitril	115 (14.6)	67 (15.9)	6 (13.3)	12 (14.0)	30 (13.6)	0 (0.0)	0.67	0.21
Diuretics	197 (46.0) (*n* = 428)	111 (48.9) (*n* = 227)	21 (70.0) (*n* = 30)	25 (47.2) (*n* = 53)	38 (36.5) (*n* = 104)	2 (14.3) (*n* = 14)	0.01	<0.01
SGLT2 inhibitors	31 (7.2) (*n* = 428)	21 (9.3) (*n* = 227)	1 (3.3) (*n* = 30)	3 (5.7) (*n* = 53)	5 (4.8) (*n* = 104)	1 (7.1) (*n* = 14)	0.76	0.15
History of VA, *n* (%)	388 (49.2)	270 (64.1)	18 (40.0)	31 (36.0)	68 (30.9)	1 (6.2)	<0.01	<0.01
VT	356 (45.2)	252 (59.9)	18 (40.0)	27 (31.4)	59 (26.8)	0 (0.0)	<0.01	<0.01

Continued

Table 1 *Continued*

	Overall population ($n = 788$)	STORM 1 ($n = 421$)	STORM 2 ($n = 45$)	STORM 3 ($n = 86$)	STORM 4 ($n = 220$)	STORM 5 ($n = 16$)	P value	P-trend
VF	36 (8.4)	25 (11.0)	0 (0.0)	3 (5.7)	8 (7.7)	0 (0.0)	0.20	0.12
ES	133 (16.9)	87 (20.7)	8 (17.8)	12 (14.0)	25 (11.4)	1 (6.2)	0.03	<0.01
VA ablation history, *n* (%)	114 (14.5)	85 (20.2)	6 (13.3)	9 (10.5)	14 (6.4)	0 (0.0)	<0.01	<0.01
ICD, *n* (%)	453 (57.5)	303 (72.0)	26 (57.8)	38 (44.2)	85 (38.6)	1 (6.2)	<0.01	<0.01
Secondary prevention	279 (35.4)	202 (48.0)	13 (28.9)	21 (24.4)	42 (19.1)	1 (6.2)	<0.01	<0.01
ES as first cardiovascular event, *n* (%)	130 (16.5)	42 (10.0)	4 (8.9)	20 (23.3)	59 (26.8)	5 (31.2)	<0.01	<0.01

ACE-I, angiotensin-converting enzyme inhibitors; ARB, angiotensin receptor blockers; BMI, body mass index; CKD, chronic kidney disease; COPD, chronic obstructive pulmonary disease; ES, electrical storm; ICD, implantable cardioverter-defibrillator; LVEF, left ventricular ejection fraction; MRA, mineralocorticoid receptor antagonist; PAD, peripheral arterial disease; S-ICD, subcutaneous implantable cardioverter defibrillator; VA, ventricular arrhythmia; VF, ventricular fibrillation; VT, ventricular tachycardia.

[a]Quantitative data are presented as median (IQR), qualitative data are presented as *n* (%). When missing data were present, the number of available observations is shown in parentheses (*n* = …).

(*Figure 4B*). Multivariate logistic regression further identified the STORM stage as one of the clinical characteristics independently associated with increased 30-day mortality (see Supplementary material online, *Table S1*). Moreover, higher STORM stages were associated with a greater incidence of both cardiovascular and non-cardiovascular death, as well as increased rates of LVAD implantation and heart transplantation (*Table 4*). Finally, a progressive increase in STORM severity was also significantly associated with higher in-hospital recurrence of VAs and complications, including infections, brain injury, pneumonia and thrombosis (but not pericardial effusion).

Subgroups analysis

The STORM classification demonstrated consistent prognostic stratification across all subgroups, with increasing mortality trend from STORM-1 to STORM-5 (*Figure 5*). In ischaemic cardiomyopathy patients, mortality ranged from 4.6% (STORM-1) to 57.1% (STORM-5), while in reduced LVEF patients it increased from 6.0% to 58.3%. Similarly, AMI patients showed preserved risk gradient from 6.8% (STORM-1) to 50.0% (STORM-5). Both monomorphic VT and polymorphic VT/VF patients maintained the same progressive mortality increase across STORM grades, confirming the robustness of this classification as a prognostic marker regardless of arrhythmia type.

Discussion

To the best of our knowledge, this study is the first to propose a multistage classification for stratifying ES severity. The main findings are as follows: (1) ES remains a severe cardiovascular condition, with an early mortality rate approaching 15% within the first month; (2) A severity-response classification framework based on four simple and clinically relevant criteria – signs of acute heart failure or haemodynamic instability; need for inotropes or vasopressors; use of advanced supportive therapies (including deep sedation) and renal replacement therapy; and implementation of mechanical circulatory support – may efficiently stratify 30-day mortality risk in patients with ES; and (3) The STORM classification demonstrated a striking and consistent escalation in 30-day mortality risk, rising from 5.0% in STORM-1 to 50.0% in STORM-5, underscoring its strong prognostic relevance.

The same name for different clinical situations

The current contemporary definition of ES is based on the occurrence of three or more separate VA within a 24-hour span, each separated by a minimum of 5 min.[1,4] These recurrent episodes are sometimes haemodynamically tolerated and managed through a comprehensive strategy integrating the use of antiarrhythmic medications,[13] ICD reprogramming,[14] correction of electrolyte imbalances,[15] and VT catheter ablation.[16] However, the clinical scenario may escalate in complexity; some patients having persistent VA despite optimal therapeutic measures associated with haemodynamic instability. Under such circumstances, standard care may involve neuraxial modulation,[17] deep sedation,[18] or mechanical circulatory support.[19] Consequently, the term 'electrical storm' applies to various clinical presentations, spanning from asymptomatic VT runs observed with remote monitoring to multiple symptomatic episodes requiring urgent medical intervention or even more critical situations with life-threatening arrhythmias. Although all are denominated as the same disease, i.e. 'electrical storm,' they vary in terms of morbidity and mortality.[20] In large studies, the overall mortality of ES has been estimated to be 14% in the first 48 h,[2] 23%[21] at 30 days and 24% to 34% at one year,[3] but one may estimate that such mortality varies widely depending on the severity of the initial presentation.

Current international guidelines emphasize the critical importance of haemodynamic assessment during ES evaluation, recommending

Table 2 Initial ES characteristics according to the STORM classification[a]

	Overall population (n = 788)	STORM 1 (n = 421)	STORM 2 (n = 45)	STORM 3 (n = 86)	STORM 4 (n = 220)	STORM 5 (n = 16)	P value	P-trend
Type of VA, n (%)								
Monomorphic VT	283 (66.1) (n = 428)	173 (76.2) (n = 227)	17 (56.7) (n = 30)	37 (69.8) (n = 53)	49 (47.1) (n = 104)	7 (50.0) (n = 14)	<0.01	<0.01
Polymorphic VT	58 (13.6) (n = 428)	26 (11.5) (n = 227)	8 (26.7) (n = 30)	3 (5.7) (n = 53)	17 (16.3) (n = 104)	4 (28.6) (n = 14	0.02	0.19
VF	153 (35.7) (n = 428)	62 (27.3) (n = 227)	6 (20.0) (n = 30)	21 (39.6) (n = 53)	56 (53.8) (n = 104)	8 (57.1) (n = 14)	<0.01	<0.01
Torsades de pointes	26 (6.1) (n = 428)	12 (5.3) (n = 227)	2 (6.7) (n = 30)	4 (7.5) (n = 53)	7 (6.7) (n = 104)	1 (7.1) (n = 14)	0.84	0.52
Therapies delivered, n (%)							<0.01	
ATP only	83 (13.7) (n = 605)	59 (17.1) (n = 345)	11 (30.6) (n = 36)	4 (6.5) (n = 62)	9 (6.1) (n = 148)	0 (0.0) (n = 14)		<0.01
Shock only	246 (40.7) (n = 605)	96 (27.8) (n = 345)	12 (33.3) (n = 36)	30 (48.4) (n = 62)	95 (64.2) (n = 148)	13 (92.9) (n = 14)		<0.01
ATP + Shock	249 (41.2) (n = 605)	171 (49.6) (n = 345)	11 (30.6) (n = 36)	26 (41.9) (n = 62)	40 (27.0) (n = 148)	1 (7.1) (n = 14)		<0.01
Blood test, median (IQR)								
Creatinine, µmol/L	115.0 (90.0–150.0) (n = 772)	110.0 (90.0–148.0) (n = 414)	102.0 (86.0–156.0) (n = 41)	120.0 (90.0–150.0) (n = 85)	121.0 (90.0–160.0) (n = 217)	118.0 (90.0–134.5) (n = 15)	0.26	0.05
Potassium, mmol/L	4.2 (3.9–4.6) (n = 767)	4.3 (3.9–4.6) (n = 414)	4.2 (3.6–4.4) (n = 40)	4.2 (3.7–4.6) (n = 84)	4.2 (3.8–4.6) (n = 214)	4.4 (4.1–4.7) (n = 15)	0.18	0.07
ES trigger, n (%)								
STEMI	117 (14.8)	25 (5.9)	7 (15.6)	15 (17.4)	57 (25.9)	13 (81.2)	<0.01	<0.01
NSTEMI	90 (11.4)	36 (8.6)	17 (37.8)	13 (15.1)	23 (10.5)	1 (6.2)	<0.01	0.49
Hypokalaemia	64 (8.1)	28 (6.7)	7 (15.6)	10 (11.6)	17 (7.7)	2 (12.5)	0.13	0.37
Hyperthyroidism	21 (2.7)	14 (3.3)	0 (0.0)	4 (4.7)	3 (1.4)	0 (0.0)	0.33	0.20
Infection	42 (5.3)	21 (5.0)	2 (4.4)	8 (9.3)	11 (5.0)	0 (0.0)	0.53	0.92
Noncompliance or treatment modification	25 (3.2)	18 (4.3)	4 (8.9)	2 (2.3)	1 (0.5)	0 (0.0)	<0.01	<0.01
Coronary angiography, n (%)	191 (44.6) (n = 428)	69 (30.4) (n = 227)	14 (46.7) (n = 30)	25 (47.2) (n = 53)	69 (66.3) (n = 104)	14 (100.0) (n = 14)	<0.01	<0.01
Revascularization	105 (24.5) (n = 428)	28 (12.3) (n = 227)	7 (23.3) (n = 30)	14 (26.4) (n = 53)	44 (42.3) (n = 104)	12 (85.7) (n = 14)	<0.01	<0.01
Fully revascularized	88 (20.6) (n = 428)	41 (18.1) (n = 227)	5 (16.7) (n = 30)	11 (20.8) (n = 53)	27 (26.0) (n = 104)	4 (28.6) (n = 14)	0.45	0.08

ATP, anti-tachycardia pacing; ES, electrical storm; NSTEMI, no ST segment elevation myocardial infarction; STEMI, ST segment elevation myocardial infarction; VA, ventricular arrhythmia; VF, ventricular fibrillation; VT, ventricular tachycardia.
[a]Quantitative data are presented as median (IQR), qualitative data are presented as n (%). When missing data were present, the number of available observations is shown in parentheses (n = …).

Table 3 ES management according to the STORM classification[a]

	Overall population ($n = 788$)	STORM 1 ($n = 421$)	STORM 2 ($n = 45$)	STORM 3 ($n = 86$)	STORM 4 ($n = 220$)	STORM 5 ($n = 16$)	*P* value	*P*-trend
Medications, *n* (%)								
Amiodarone	662 (84.0)	330 (78.4)	38 (84.4)	77 (89.5)	203 (92.3)	14 (87.5)	<0.01	<0.01
Lidocaine	349 (44.3)	122 (29.0)	16 (35.6)	45 (52.3)	154 (70.0)	12 (75.0)	<0.01	<0.01
Betablocker	304 (71.0) ($n = 428$)	181 (79.7) ($n = 227$)	23 (76.7) ($n = 30$)	34 (64.2) ($n = 53$)	59 (56.7) ($n = 104$)	7 (50.0) ($n = 14$)	<0.01	<0.01
Isoprenaline	10 (1.3)	5 (1.2)	1 (2.2)	2 (2.3)	2 (0.9)	0 (0.0)	0.52	0.83
Magnesium sulphate	150 (35.0) ($n = 428$)	73 (32.2) ($n = 227$)	11 (36.7) ($n = 30$)	24 (45.3) ($n = 53$)	36 (34.6) ($n = 104$)	6 (42.9) ($n = 14$)	0.43	0.29
Mexiletine	16 (3.7) ($n = 428$)	12 (5.3) ($n = 227$)	0 (0.0) ($n = 30$)	2 (3.8) ($n = 53$)	2 (1.9) ($n = 104$)	0 (0.0) ($n = 14$)	0.55	0.10
Inotropes/vasopressors								
Dobutamine	186 (23.6)	0 (0.0)	0 (0.0)	32 (37.2)	139 (63.2)	15 (93.8)	<0.01	<0.01
Norepinephrine	201 (25.5)	0 (0.0)	0 (0.0)	45 (52.3)	142 (64.5)	14 (87.5)	<0.01	<0.01
Epinephrine	44 (5.6)	0 (0.0)	0 (0.0)	9 (10.5)	27 (12.3)	8 (50.0)	<0.01	<0.01
Light sedation, *n* (%)	117 (27.3) ($n = 428$)	58 (25.6) ($n = 227$)	10 (33.3) ($n = 30$)	20 (37.7) ($n = 53$)	26 (25.0) ($n = 104$)	3 (21.4) ($n = 14$)	0.36	0.86
Neuraxial modulation, *n* (%)	23 (2.9)	4 (1.0)	1 (2.2)	5 (5.8)	9 (4.1)	4 (25.0)	<0.01	<0.01
Deep sedation, *n* (%)	233 (29.6)	0 (0.0)	0 (0.0)	44 (51.2)	175 (79.5)	14 (87.5)	<0.01	<0.01
Renal replacement therapy, *n* (%)	36 (4.6)	0 (0.0)	0 (0.0)	3 (3.5)	28 (12.7)	5 (31.2)	<0.01	<0.01
IABP, *n* (%)	21 (2.7)	0 (0.0)	0 (0.0)	0 (0.0)	10 (4.5)	11 (68.8)	<0.01	<0.01
Impella, *n* (%)	12 (1.5)	0 (0.0)	0 (0.0)	0 (0.0)	5 (2.3)	7 (43.8)	<0.01	<0.01
ECMO, *n* (%)	80 (10.2)	0 (0.0)	0 (0.0)	0 (0.0)	65 (29.5)	15 (93.8)	<0.01	<0.01
Catheter ablation, *n* (%)	284 (36.0)	182 (43.2)	14 (33.9)	28 (32.6)	59 (26.8)	1 (6.2)	<0.01	<0.01
Stereotactic ablation, *n* (%)	22 (2.8)	17 (4.0)	3 (6.7)	1 (1.2)	1 (0.5)	0 (0.0)	0.02	<0.01

ES, electrical storm; ECMO, extracorporeal membrane oxygenation; IABP, intra-aortic balloon pump.
[a]Quantitative data are presented as median (IQR), qualitative data are presented as *n* (%). When missing data were present, the number of available observations is shown in parentheses ($n = \ldots$).

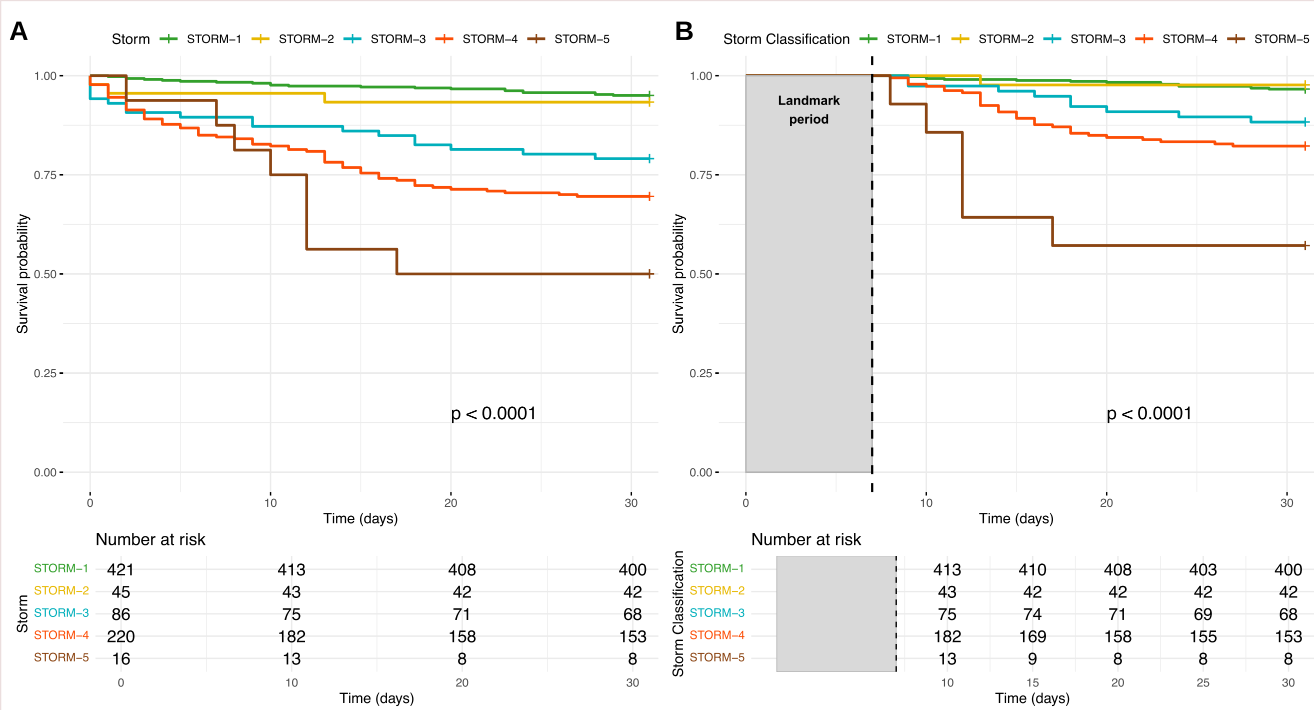


Figure 4 Kaplan–Meier survival curves for 30-day mortality stratified by STORM classification. (*A*) Complete analysis showing all patients from day 0 to 30. (*B*) Landmark analysis at day 7 demonstrating prognostic discrimination after therapeutic stabilization period.

identification of coexisting decompensated heart failure or cardiogenic shock and assessment of perfusion status.[4,5] However, these guidelines do not provide a standardized classification framework to operationalize this assessment or stratify mortality risk. Thus, there is a critical need for a clinically grounded classification system to better define the risk profile of patients admitted for ES and to guide therapeutic decision-making. To this end, based on recent studies and expert consensus[1,3,5,12] we developed a severity-based staging model by systematically analysing four key clinical parameters reflecting both disease severity and treatment intensity: (1) signs of acute heart failure or haemodynamic instability requiring urgent intervention, which may independently worsen prognosis regardless of underlying etiology[3]; (2) the need for inotropes or vasopressors, as a marker of peripheral hypoperfusion[4]; (3) the requirement for advanced supportive therapies (including deep sedation) or renal replacement therapy to manage refractory arrhythmias, both associated with poor outcomes[18]; and (4) the use of MCS devices, which have consistently been linked to increased mortality.[19] The clinical response to the initial first therapeutic steps is critical in determining whether the situation is going to be easy to handle. Based on these clinical characteristics, five stages of progressive clinical complexity were created. As described in this study, a progressive increase in mortality was observed from benign to more severe forms of ES, ranging from 5% to stage 1 up to 50% in stage 5. Jentzer et al.[4] recently highlighted the importance of a rational stepwise approach for medical therapy in ES, individualizing patients at low and at high-risk, based on the haemodynamic tolerance, the number of VA episodes and the therapies delivered. This approach aimed to adapt ES management to the clinical presentation and anticipate the need for haemodynamic support. Here, we aimed to create an easy-to-use severity-response framework capable of predicting ES mortality, leveraging basic clinical data and validated treatments for ES and cardiogenic shock. This represents a significant advance over current practice, where patients with ES are managed without standardized severity assessment, limiting both therapeutic decision-making and the ability to compare outcomes across studies.

Clinical implications

As the first severity classification system specifically designed for patients with ES, the STORM framework addresses a critical gap identified in current guidelines.[4,5] If externally validated, it would help multidisciplinary teams managing ES to stratify patient risk and predict 30-day mortality. Whether catheter ablation provides a clear benefit in terms of outcomes and optimal timing remains to be determined in future studies.[22] Additionally, ES – whether due to its acute severity or its frequent occurrence in the context of advanced cardiomyopathy, often reflecting a progression of myocardial decline – may serve as a red flag for heart transplantation.[23] However, significant uncertainty persists regarding the optimal criteria to guide timely referral of patients with ES for advanced heart failure therapies.[24] In this context, the maximum STORM stage attained during hospitalization may serve as a pragmatic severity marker to inform patient selection and guide management strategies, including consideration for transplant evaluation. Beyond this retrospective application, the STORM classification could also be applied dynamically during hospitalization, allowing serial reassessment as the patient's condition evolves. Since STORM staging relies on readily available clinical and therapeutic parameters, real-time updates would enable clinicians to track clinical deterioration and potentially trigger earlier involvement of advanced heart failure teams when transitions to higher stages occur despite initial interventions.

Importantly, the STORM classification addresses a critical gap: standardized severity assessment to guide resource deployment in ES management. While STORM does not dictate specific antiarrhythmic therapeutic choices – which depend on arrhythmia mechanism and substrate characteristics – it provides the severity stratification

Table 4 In-hospital and long-term outcomes according to the STORM classification[a]

	Overall population ($n = 788$)	STORM 1 ($n = 421$)	STORM 2 ($n = 45$)	STORM 3 ($n = 86$)	STORM 4 ($n = 220$)	STORM 5 ($n = 16$)	*P* value	*P*-trend
In-hospital VA recurrence, *n* (%)								
Overall	145 (33.9) ($n = 428$)	62 (27.3) ($n = 227$)	10 (33.3) ($n = 30$)	21 (39.6) ($n = 53$)	44 (42.3) ($n = 104$)	8 (57.1) ($n = 14$)	0.02	<0.01
ES	46 (10.7) ($n = 428$)	13 (5.7) ($n = 227$)	2 (6.7) ($n = 30$)	9 (17.0) ($n = 53$)	18 (17.3) ($n = 104$)	4 (28.6) ($n = 14$)	<0.01	<0.01
VT	111 (25.9) ($n = 428$)	56 (24.7) ($n = 227$)	9 (30.0) ($n = 30$)	11 (20.8) ($n = 53$)	30 (28.8) ($n = 104$)	5 (35.7) ($n = 14$)	0.64	0.41
FV	22 (5.1) ($n = 428$)	5 (2.2) ($n = 227$)	1 (3.3) ($n = 30$)	2 (3.8) ($n = 53$)	12 (11.5) ($n = 104$)	2 (14.3) ($n = 14$)	<0.01	<0.01
In-hospital complications, *n* (%)								
Infection	56 (13.1) ($n = 428$)	8 (3.5) ($n = 227$)	2 (6.7) ($n = 30$)	8 (15.1) ($n = 53$)	27 (26.0) ($n = 104$)	11 (78.6) ($n = 14$)	<0.01	<0.01
Brain injury	25 (5.8) ($n = 428$)	3 (1.3) ($n = 227$)	0 (0.0) ($n = 30$)	2 (3.8) ($n = 53$)	17 (16.3) ($n = 104$)	3 (21.4) ($n = 14$)	<0.01	<0.01
Pneumopathy	50 (11.7) ($n = 428$)	1 (0.4) ($n = 227$)	0 (0.0) ($n = 30$)	11 (20.8) ($n = 53$)	33 (31.7) ($n = 104$)	5 (35.7) ($n = 14$)	<0.01	<0.01
Thrombosis	19 (4.4) ($n = 428$)	7 (3.1) ($n = 227$)	0 (0.0) ($n = 30$)	2 (3.8) ($n = 53$)	9 (8.7) ($n = 104$)	1 (7.1) ($n = 14$)	0.13	0.03
Pericardial effusion	6 (1.4) ($n = 428$)	4 (1.8) ($n = 227$)	0 (0.0) ($n = 30$)	1 (1.9) ($n = 53$)	1 (1.0) ($n = 104$)	0 (0.0) ($n = 14$)	1.00	0.54
LVAD, *n* (%)	21.0 (2.7)	2 (0.5)	1 (2.2)	3 (3.5)	12 (5.5)	3 (18.8)	<0.01	<0.01
Heart transplantation, *n* (%)	26 (3.3) ($n = 788$)	7 (1.7)	1 (2.2)	2 (2.3)	15 (6.8)	1 (6.2)	<0.01	<0.01
Causes of death, *n* (%)								
Cardiovascular	87 (11.0)	13 (3.1)	2 (4.4)	13 (15.1)	53 (24.1)	6 (37.5)	<0.01	<0.01
ES	41 (5.2)	9 (2.1)	2 (4.4)	6 (7.0)	22 (10.0)	2 (12.5)	<0.01	<0.01
End-stage HF	47 (6.0)	5 (1.2)	0 (0.0)	8 (9.3)	30 (13.6)	4 (25.0)	<0.01	<0.01
Other	7 (0.9)	0 (0.0)	0 (0.0)	2 (2.3)	5 (2.3)	0 (0.0)	0.01	<0.01
Non-cardiovascular	32 (4.1)	1 (0.2)	0 (0.0)	6 (7.0)	22 (10.0)	3 (18.8)	<0.01	<0.01
Infection	11 (1.4)	1 (0.2)	0 (0.0)	2 (2.3)	8 (3.6)	0 (0.0)	<0.01	<0.01
Brain death	14 (1.8)	0 (0.0)	0 (0.0)	3 (3.5)	8 (3.6)	3 (18.8)	<0.01	<0.01
Multiple organ failure	17 (2.2)	0 (0.0)	0 (0.0)	2 (2.3)	15 (6.8)	0 (0.0)	<0.01	<0.01
Unknown	6 (0.8)	1 (0.2)	0 (0.0)	2 (2.3)	3 (1.4)	0 (0.0)	0.16	0.08

ES, electrical storm; HF, heart failure; LVAD, left ventricular assist device; VA, ventricular arrhythmia; VF, ventricular fibrillation; VT, ventricular tachycardia.
[a]Quantitative data are presented as median (IQR), qualitative data are presented as *n* (%). When missing data were present, the number of available observations is shown in parentheses ($n = \ldots$).

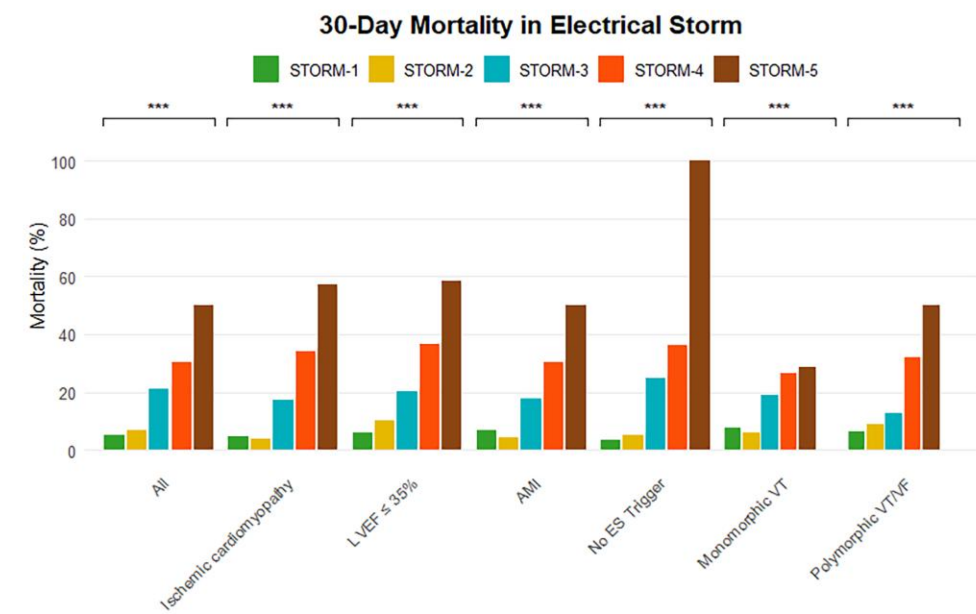


Figure 5 30-day mortality rates by STORM classification across clinically relevant subgroups. *** $P < 0.01$. AMI included both STEMI and NSTEMI triggers. ES, electrical storm; LVEF, left ventricular ejection fraction; VF, ventricular fibrillation; VT, ventricular tachycardia.

framework necessary for clinical decision-making regarding monitoring intensity, timing of advanced heart failure team involvement and standardization of ES severity reporting across research studies. This fills a void explicitly identified in current guidelines, which recommend haemodynamic evaluation without providing an operational framework.[1,4,5] Furthermore, none of the studies on ES published thus far have precisely described the type of storm included, being potentially benign storms (corresponding to stage 1), or more severe and life-threatening situations (stage 4 or 5). Consequently, analysing the results of a given ES study and comparing the results between studies is not an easy task. For instance, in the study by Chatzidou et al.[25] comparing propranolol to metoprolol in addition to amiodarone for ES, only 5 of the 60 included patients experienced what the authors defined as severe ES – characterized by haemodynamic instability, pulmonary oedema and persistent arrhythmia requiring emergent care with inotropes and haemodynamic support. Consequently, one may think that most of the included patients had stages 1 and 2 according to the current classification, and that the results of that study would not be applicable to more severe ES. Generalizing the classification described in this manuscript would then help better characterize the population included and the severity of the clinical presentation of the patients.

The STORM classification shares conceptual similarities with the Society for Cardiovascular Angiography and Interventions (SCAI) shock staging system, which stratifies cardiogenic shock severity from Stage A (at risk) to Stage E (extremis) based on clinical and haemodynamic parameters.[26] However, STORM addresses a fundamentally different clinical entity: SCAI was designed for cardiogenic shock with progressive pump failure, whereas ES is primarily an electrical disorder – 53.4% of our cohort remained haemodynamically stable (STORM-1) yet required specialized management. Critically, SCAI staging requires systematic invasive haemodynamic monitoring (cardiac index, wedge pressure, cardiac power output via pulmonary artery catheterization),[26] which is neither feasible nor warranted in most ES patients and unavailable in retrospective ES studies. Furthermore, STORM incorporates ES-specific therapeutic interventions such as deep sedation as an anti-arrhythmic strategy (used in 29.6% of our cohort) that reflect the unique pathophysiology of ES, characterized by sudden electrical deterioration rather than progressive pump failure.

Limitations

We acknowledge some limitations in our study. Our analysis was a retrospective review of a cohort of patients hospitalized in the context of ES with the inherent limitations of such studies. The classification described in this manuscript was not built based on clinical, biological, or echocardiographic parameters significantly associated with mortality in our population; however, ES characteristics were subjectively selected a priori for their perceived impact on patients' outcomes. Therefore, this choice is undoubtedly critical. For instance, neither LVEF nor arrhythmia cycle length was integrated into the classifications, although one may assume that they would impact in-hospital outcomes. However, the use of proxies reflecting the extent of global hypoperfusion, such as the need for inotropes or vasopressors, may provide a more integrative assessment of the multiple factors influencing haemodynamic tolerance and overall severity. Similarly, arrhythmia burden was not integrated because, owing to the retrospective nature of the study, this parameter was not available for all patients. However, this may also be the case in some instances in clinical practice. Moreover, VA catheter ablation was performed in only 36.0% of patients. Whether a broader application of ablation could have influenced 30-day mortality remains uncertain and warrants further investigation. STORM-1 patients were generally older and presented with a higher burden of comorbidities, which may have led to an underutilization of inotropes, vasopressors and MCS due to early decisions regarding withdrawal of life-sustaining therapies, potentially introducing a self-fulfilling prophecy bias. However, this bias would rather lead to an overestimation of mortality in STORM-1 and STORM-2 patients, and given the clear and progressive increase in mortality across STORM stages, its overall impact on the classification's prognostic validity appears limited.

Another limitation is that the STORM classification was applied retrospectively, assigning each patient the highest stage reached during hospitalization. By design, this approach conflates illness severity with treatment intensity, since parameters such as vasopressor use, renal replacement therapy, or mechanical circulatory support are both markers and consequences of clinical deterioration. However, these stratification elements were deliberately used as pragmatic proxies, given the retrospective design of the study and the lack of repeated haemodynamic or perfusion markers such as lactate levels or mottling scores. In addition, ES may deteriorate very abruptly despite similar baseline characteristics, further supporting the use of treatment escalation as a surrogate for evolving severity. Rather than serving as a purely predictive tool at admission, the STORM classification was intended as a pragmatic severity staging system reflecting both the biological aggressiveness of ES and the therapeutic escalation it necessitates. In clinical practice, however, patient status may evolve rapidly, and a dynamic re-evaluation of STORM stage might provide more accurate real-time risk stratification. Future studies should investigate the potential value of such a dynamic approach, capturing transitions between stages to better reflect the clinical trajectory of ES patients. Finally, this classification lacks external validation and should be tested in larger prospective studies to confirm its ability to predict short-term mortality after ES.

Conclusion

ES remains a severe cardiovascular event, with approximately 15% of patients dying within the first 30 days. As the first severity-response classification system specifically designed for ES, the STORM classification offers a simple and effective framework to categorize patients by clinical complexity, may facilitate standardized multidisciplinary management strategies, and effectively stratifies in-hospital mortality risk, ranging from 5% in stage 1% to 50% in stage 5. Further prospective studies are warranted to validate these findings and confirm its clinical utility.

Supplementary material

Supplementary material is available at *European Heart Journal: Acute Cardiovascular Care* online.

Author contributions

Raphaël Martins (Conceptualization [lead]; Data curation [lead]; Project administration [lead]; Supervision [lead]; Validation [lead]; Visualization [lead]; Writing – original draft [lead]; Writing – review & editing [lead]), Christophe Leclercq (Writing – review & editing [equal]), Clément Delmas (Writing – review & editing [equal]), Dominique Pavin (Data curation [lead]; Writing – review & editing [equal]), Antoine Da Costa (Writing – review & editing [equal]), Rayan Mohamed (Data curation [lead]; Writing – review & editing [equal]), Lucas Deville (Data curation [lead]; Writing – review & editing [equal]), Gabriel Laurent (Writing – review & editing [equal]), Soundous M'Rabet (Data curation [lead]; Writing – review & editing [equal]), Alexandre Salaun (Data curation [equal]; Writing – review & editing [equal]), Cédric Klein (Data curation [lead]; Writing – review & editing [equal]), Amine Tazibet (Data curation [lead]; Writing – review & editing [equal]), Pierre Groussin (MD (Data curation [lead])), Donovan Decaudin (Data curation [lead]; Writing – review & editing [equal]), Karim Benali (MD PhD (Conceptualization [lead]; Data curation [lead]; Methodology [lead]; Supervision [lead]; Writing – review & editing [lead])), Miloud CHERBI (MD), Charles Guenancia (Conceptualization [lead]; Data curation [lead]; Supervision [lead]; Writing – review & editing [equal]), and Sandro Ninni (Conceptualization [lead]; Data curation [lead]; Project administration [lead]; Supervision [lead]; Writing – review & editing [lead])

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Data availability

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

References

1. Zeppenfeld K, Tfelt-Hansen J, de Riva M, Winkel BG, Behr ER, Blom NA, *et al.* 2022 ESC guidelines for the management of patients with ventricular arrhythmias and the prevention of sudden cardiac death. *Eur Heart J* 2022;**43**:3997–4126.
2. Geraghty L, Santangeli P, Tedrow UB, Shivkumar K, Kumar S. Contemporary management of electrical storm. *Heart Lung Circ* 2019;**28**:123–133.
3. Ninni S, Layec J, Brigadeau F, Behal H, Labreuche J, Klein C, *et al.* Incidence and predictors of mortality after an electrical storm in the ICU. *Eur Heart J Acute Cardiovasc Care* 2022; **11**:431–439.
4. Jentzer JC, Noseworthy PA, Kashou AH, May AM, Chrispin J, Kabra R, *et al.* Multidisciplinary critical care management of electrical storm: JACC state-of-the-art review. *J Am Coll Cardiol* 2023;**81**:2189–2206.
5. Lenarczyk R, Zeppenfeld K, Tfelt-Hansen J, Heinzel FR, Deneke T, Ene E, *et al.* Management of patients with an electrical storm or clustered ventricular arrhythmias: a clinical consensus statement of the European Heart Rhythm Association of the ESC-endorsed by the Asia-Pacific Heart Rhythm Society, Heart Rhythm Society, and Latin-American Heart Rhythm Society. *Europace* 2024;**26**:euae049.
6. Vaseghi M, Barwad P, Malavassi Corrales FJ, Tandri H, Mathuria N, Shah R, *et al.* Cardiac sympathetic denervation for refractory ventricular arrhythmias. *J Am Coll Cardiol* 2017; **69**:3070–3080.
7. Guerra F, Palmisano P, Dell'Era G, Ziacchi M, Ammendola E, Bonelli P, *et al.* Implantable cardioverter-defibrillator programming and electrical storm: results of the OBSERVational registry on long-term outcome of ICD patients (OBSERVO-ICD). *Heart Rhythm* 2016;**13**:1987–1992.
8. Elsokkari I, Parkash R, Tang A, Wells G, Doucette S, Yetisir E, *et al.* Mortality risk increases with clustered ventricular arrhythmias in patients with implantable cardioverter-defibrillators. *JACC Clin Electrophysiol* 2020;**6**:327–337.
9. Ninni S, Guenancia C, Bourdrel O, Mohammed R, Decaudin D, Klein C, *et al.* Impact of ventricular tachycardia ablation in the setting of electrical storm based on patient risk profile. *Europace* 2025;**27**:euaf188.
10. Salaun A, Martins R, Ninni S, Mohammed R, Sava R, Decaudin D, *et al.* Characteristics and prognosis of electrical storm during acute myocardial infarction. *Europace* 2025; **27**:euaf077.
11. McDonagh TA, Metra M, Adamo M, Gardner RS, Baumbach A, Böhm M, *et al.* 2021 ESC guidelines for the diagnosis and treatment of acute and chronic heart failure. *Eur Heart J* 2021;**42**:3599–3726.
12. Incognito C, Martens P, Hedley J, Parker JD, Posadas K, Gangidi S, *et al.* Predictors and outcome of electrical storm-induced cardiogenic shock. *Eur Heart J Acute Cardiovasc Care* 2022;**11**:906–915.
13. Connolly SJ, Dorian P, Roberts RS, Gent M, Bailin S, Fain ES, *et al.* Comparison of beta-blockers, amiodarone plus beta-blockers, or sotalol for prevention of shocks from implantable cardioverter defibrillators: the OPTIC study: a randomized trial. *JAMA* 2006; **295**:165–171.
14. Moss AJ, Schuger C, Beck CA, Brown MW, Cannom DS, Daubert JP, *et al.* Reduction in inappropriate therapy and mortality through ICD programming. *N Engl J Med* 2012;**367**: 2275–2283.
15. Hendriks AA, Szili-Torok T. Editor's choice-the treatment of electrical storm: an educational review. *Eur Heart J Acute Cardiovasc Care* 2018;**7**:478–483.
16. Benali K, Ninni S, Guenancia C, Mohammed R, Decaudin D, Bourdrel O, *et al.* Impact of catheter ablation of electrical storm on survival: a propensity score-matched analysis. *JACC Clin Electrophysiol* 2024;**10**:2117–2128.
17. Savastano S, Baldi E, Compagnoni S, Rordorf R, Sanzo A, Gentile FR, *et al.* Electrical storm treatment by percutaneous stellate ganglion block: the STAR study. *Eur Heart J* 2024;**45**:823–833.
18. Martins RP, Urien JM, Barbarot N, Rieul G, Sellal JM, Borella L, *et al.* Effectiveness of deep sedation for patients with intractable electrical storm refractory to antiarrhythmic drugs. *Circulation* 2020;**142**:1599–1601.
19. Durães-Campos I, Costa C, Ferreira AR, Basílio C, Torrella P, Neves A, *et al.* ECMO for drug-refractory electrical storm without a reversible trigger: a retrospective multicentric observational study. *ESC Heart Fail* 2024;**11**:2129–2137.
20. Samuel M, Elsokkari I, Sapp JL. Ventricular tachycardia burden and mortality: association or causality? *Can J Cardiol* 2022;**38**:454–464.

21. Jiménez Candil J, Castro JC, Hernández J, Fernández-Portales J, Durán M, Durán O, *et al.* Timing of ablation and prognosis of patients with electrical storm and scar-related left ventricular dysfunction. *Am J Cardiol* 2020;**136**:87–93.
22. Lin AY, Begur M, Margolin E, Brann A, Ho G, Han F, *et al.* Catheter ablation vs advanced therapy for patients with severe heart failure and ventricular electrical storm. *Heart Rhythm* 2025;**22**:693–700.
23. Martins RP, Hamel-Bougault M, Bessière F, Pozzi M, Extramiana F, Brouk Z, *et al.* Heart transplantation as a rescue strategy for patients with refractory electrical storm. *Eur Heart J Acute Cardiovasc Care* 2023;**12**:571–581.
24. Cherbi M, Maury P, Benabou L, Faure M, Pozzi M, Sebbag L, *et al.* Long-term outcomes of electrical storm patients listed for urgent heart transplantation but not transplanted acutely. *ESC Heart Fail* 2025;**12**:1583–1593.
25. Chatzidou S, Kontogiannis C, Tsilimigras DI, Georgiopoulos G, Kosmopoulos M, Papadopoulou E, *et al.* Propranolol versus metoprolol for treatment of electrical storm in patients with implantable cardioverter-defibrillator. *J Am Coll Cardiol* 2018;**71**:1897–1906.
26. Naidu SS, Baran DA, Jentzer JC, Hollenberg SM, van Diepen S, Basir MB, *et al.* SCAI SHOCK stage classification expert consensus update: a review and incorporation of validation studies: this statement was endorsed by the American College of Cardiology (ACC), American College of Emergency Physicians (ACEP), American Heart Association (AHA), European Society of Cardiology (ESC) Association for Acute Cardiovascular Care (ACVC), International Society for Heart and Lung Transplantation (ISHLT), Society of Critical Care Medicine (SCCM), and Society of Thoracic Surgeons (STS) in December 2021. *J Am Coll Cardiol* 2022;**79**:933–946.